

Lab Title: P2P Data Leakage: Insider Theft and Torrent Distribution

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Course Name: Computer Forensics Case Study I (CIP-B104)

Date of Submission: 9/3/2026

Instruction Name: Aminu Idris

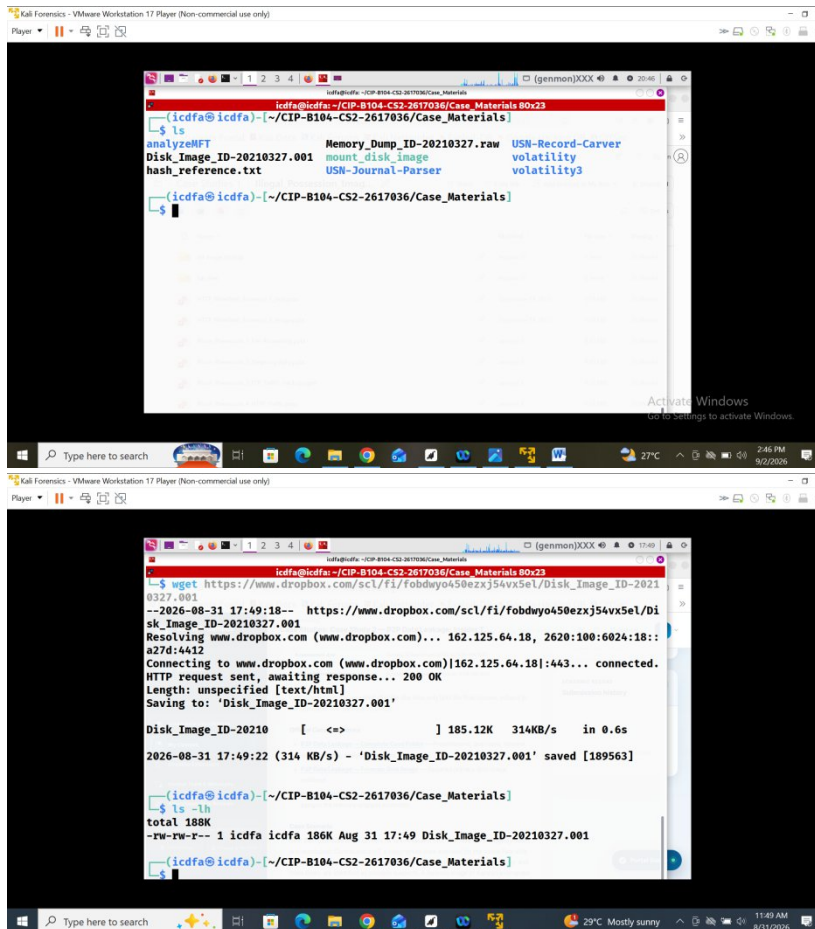
Platform Used: ICDFA Kali Linux

Executive Summary

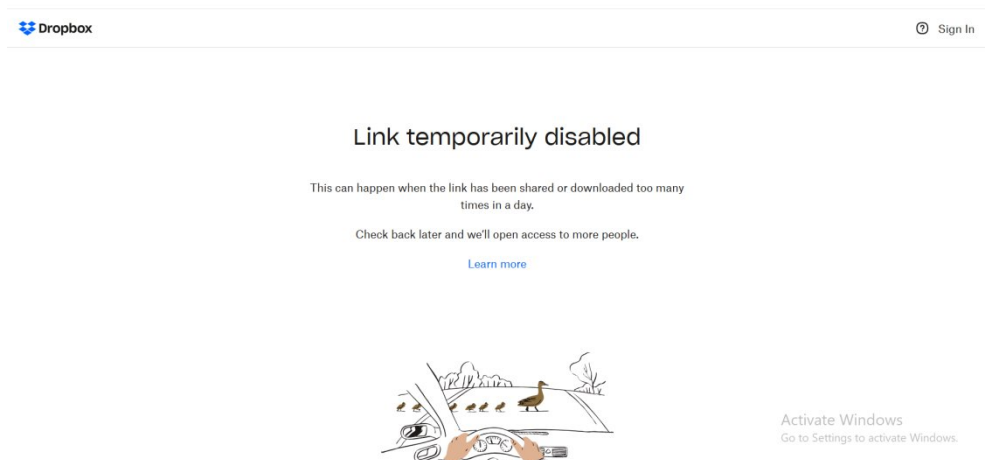
This investigation examined the alleged theft and distribution of Beat Step's Sample-1.mp3 and Contraband.mp3 files. The forensic disk image could not be downloaded because the provided link was temporarily disabled, limiting full evidence analysis and verification. However, relevant forensic commands and techniques were performed in the lab using the available case materials. The investigation focused on file, torrent, email, browser, and timeline evidence to identify the source and potential individuals associated with the leakage.

The Lab Structure

1. Here I created the structure for the lab, and I downloaded the image file to be worked on but for this lab I was unable to download the image file as I keep getting expired link



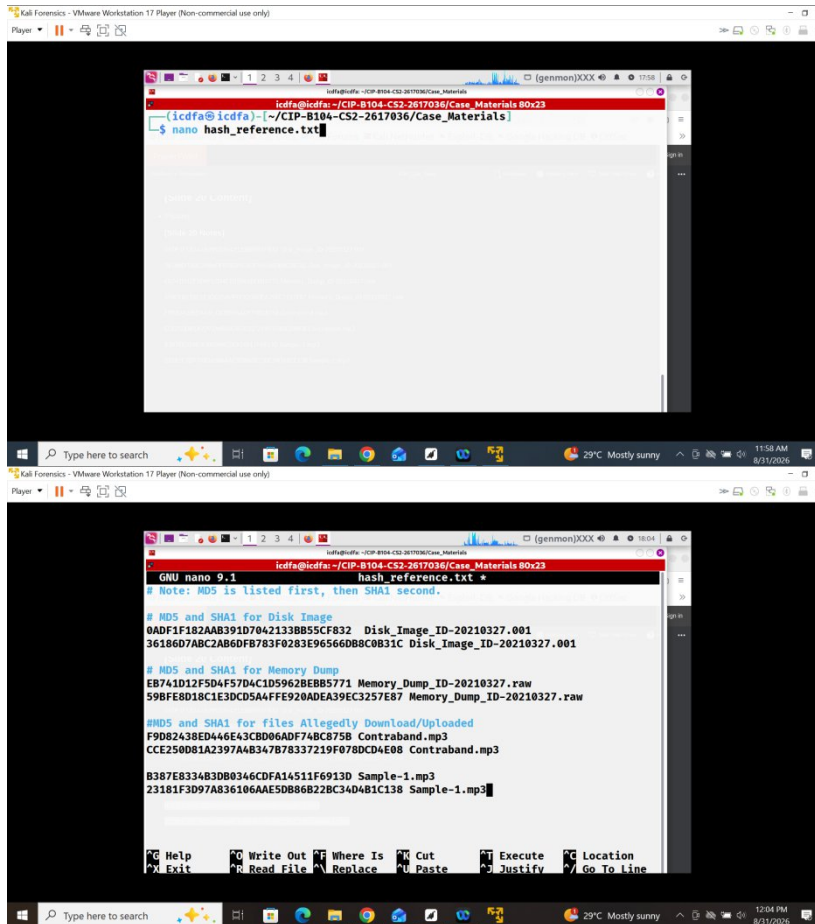
Here is the evidence of the expired link



I didn't allow that stop me and I still try more commands from the lab.

The Text File

1. I created the nano.reference.txt file and added the stated values from the lab



The first screenshot shows a terminal window with the command `nano hash_reference.txt` being executed. The second screenshot shows the contents of the `hash_reference.txt` file being edited in nano. The file contains MD5 and SHA1 hashes for a disk image, a memory dump, and two MP3 files.

```
icdfa@icdfa:~/CIP-B104-CS2-2617036/Case_Materials Box23
icdfa@icdfa:~/CIP-B104-CS2-2617036/Case_Materials
$ nano hash_reference.txt

GNU nano 9.1 hash_reference.txt
# Note: MD5 is listed first, then SHA1 second.

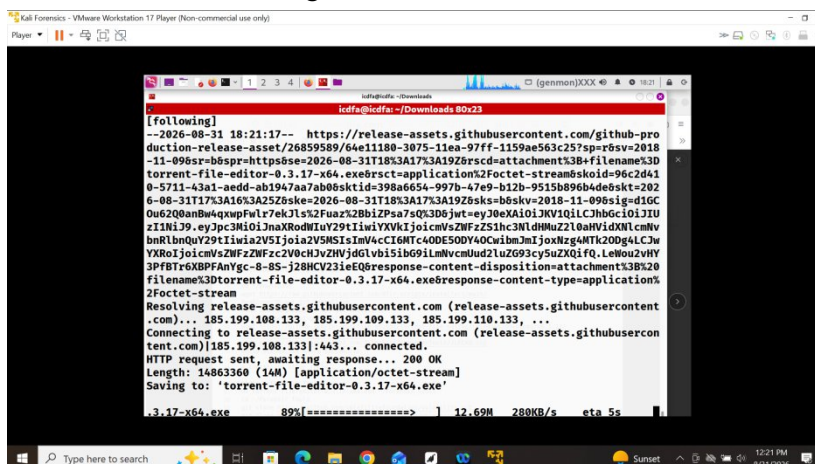
# MD5 and SHA1 for Disk Image
0ADF1F182AAB391D70421338B55CF832 Disk_Image_ID-20210327.001
3618607ABC2AB60FB783F0283E965660B8C0B31C Disk_Image_ID-20210327.001

# MD5 and SHA1 for Memory Dump
EB741D12F5D4F57D4C1D59628EBB5771 Memory_Dump_ID-20210327.raw
59BF8B8D18C1E3DCD5A4FFE920ADEA39EC3257E87 Memory_Dump_ID-20210327.raw

#MD5 and SHA1 for files Allegedly Download/Uploaded
F9D82438ED446E43C8D06ADF7ABC875B Contraband.mp3
CCE25081A2397A4B347B78337219F078DCD4E88 Contraband.mp3

B387E8334B3DB0346CDFA14511F6913D Sample-1.mp3
23181F3D97A836106AAE5DB86B2B34D481C138 Sample-1.mp3
```

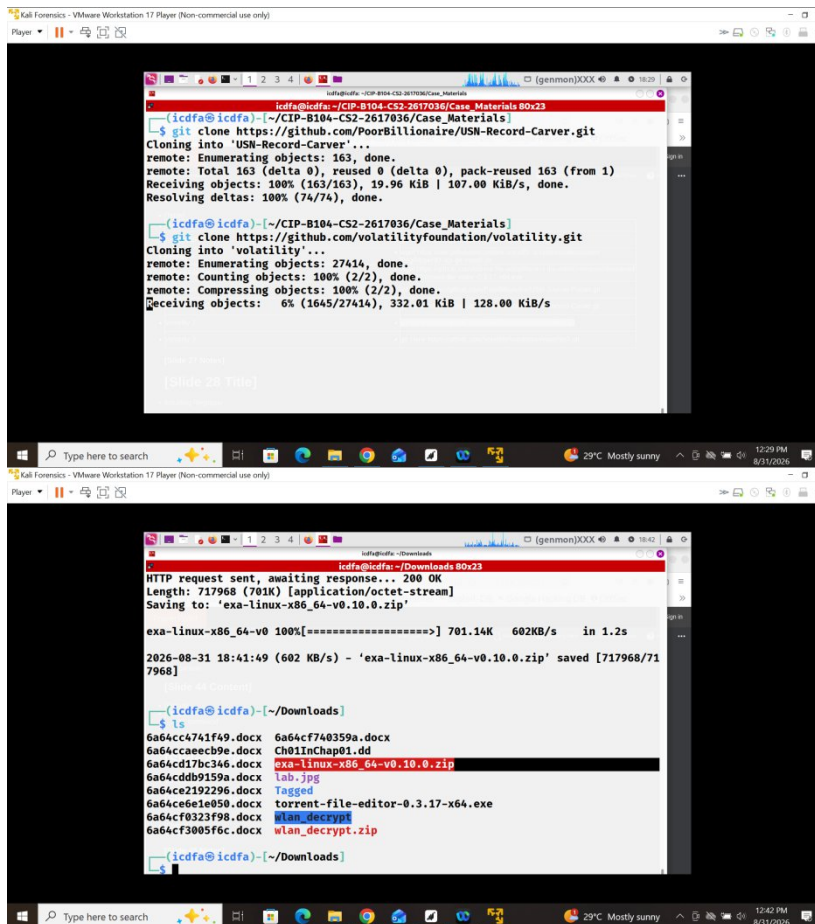
2. Here I was downloading some software commands to be used in the lab



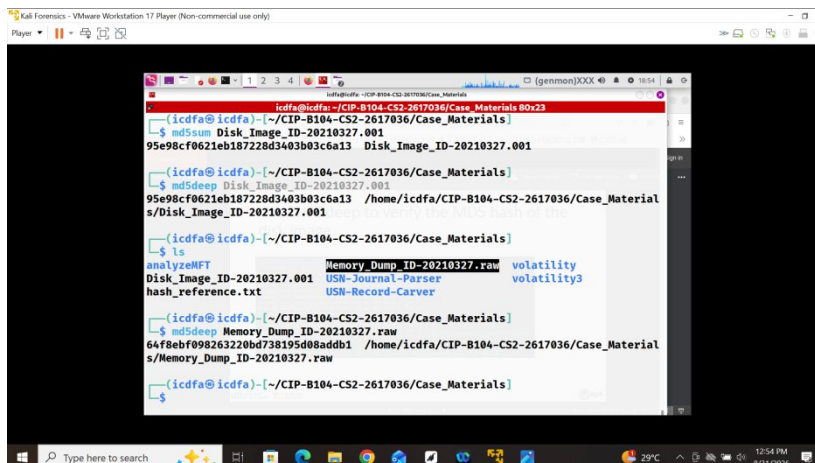
The screenshot shows a terminal window with the command `wget https://release-assets.githubusercontent.com/github-production-release-asset/26859589/64e11180-3075-11ea-97ff-1159ae563c25?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-08-31T18%3A17%3A19Z&rs=cd=attachment%3B+filename%3Dtorrent-file-editor-0.3.17-x64.exe&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-08-31T17%3A16%3A25Z&sk=2026-08-31T18%3A17%3A19Z&sk=3b6skv-2018-11-09&sig=d16C0u62QanBw&exp=Full7ekJLs2Fuaz2Z8b1ZPa7sQk306jey30eXAI0iJkV10i1CJhb6c101J1Uz11Ni39_eyJpc3MiOiJnaXRodWluY29tIiwia2V5MSIsImV4cCI6MTY4MDU0ODU0CiwibmJmJjoxNzg4MTk2ODg4LjJwYXR0IjoicmVsZWZzZWZzZWVhZGlvdjB1bG9iLmNvcmlud2luZG93cy5uZXQ1fQ.LeWou2vHY3PFBT76XBPfAnYgc-8-85-j28HCv231eEQ&response-content-disposition=attachment%3B%20filename%3Dtorrent-file-editor-0.3.17-x64.exe&response-content-type=application%2Foctet-stream` being executed. The output shows the file being downloaded and saved as `torrent-file-editor-0.3.17-x64.exe`.

```
icdfa@icdfa:~/Downloads Box23
[following]
--2026-08-31 18:21:17-- https://release-assets.githubusercontent.com/github-pro
duction-release-asset/26859589/64e11180-3075-11ea-97ff-1159ae563c25?sp=r&sv=2018
-11-09&sr=b&spr=https&se=2026-08-31T18%3A17%3A19Z&rs=cd=attachment%3B+filename%3D
torrent-file-editor-0.3.17-x64.exe&rsct=application%2Foctet-stream&skoid=96c2d41
0-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=202
6-08-31T17%3A16%3A25Z&sk=2026-08-31T18%3A17%3A19Z&sk=3b6skv-2018-11-09&sig=d16C
0u62QanBw&exp=Full7ekJLs2Fuaz2Z8b1ZPa7sQk306jey30eXAI0iJkV10i1CJhb6c101J1U
z11Ni39_eyJpc3MiOiJnaXRodWluY29tIiwia2V5MSIsImV4cCI6MTY4MDU0ODU0CiwibmJmJjoxNzg4MTk2ODg4LjJw
YXR0IjoicmVsZWZzZWZzZWVhZGlvdjB1bG9iLmNvcmlud2luZG93cy5uZXQ1fQ.LeWou2vHY
3PFBT76XBPfAnYgc-8-85-j28HCv231eEQ&response-content-disposition=attachment%3B%20
filename%3Dtorrent-file-editor-0.3.17-x64.exe&response-content-type=application%
2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent
.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercon
tent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14863360 (14M) [application/octet-stream]
Saving to: 'torrent-file-editor-0.3.17-x64.exe'

.3.17-x64.exe      89%[=====>] 12.69M  280KB/s  eta 5s
```

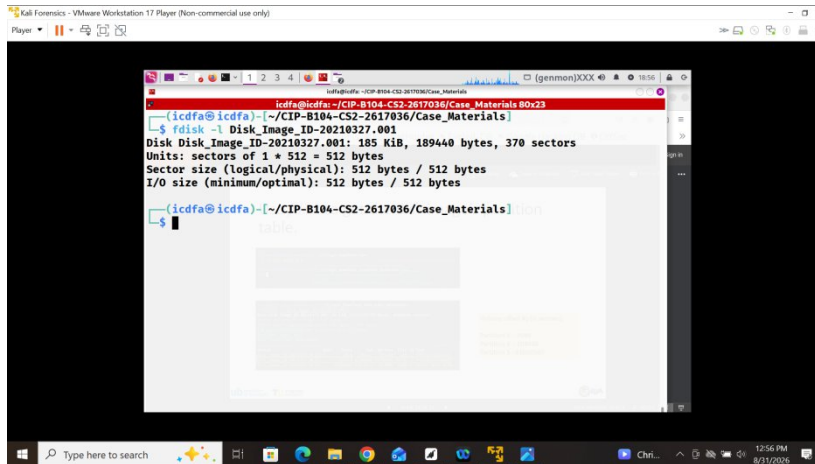


3. I tried pulling the hash value image from the disk image I downloaded but it did not tally



Disk Image and Partitions

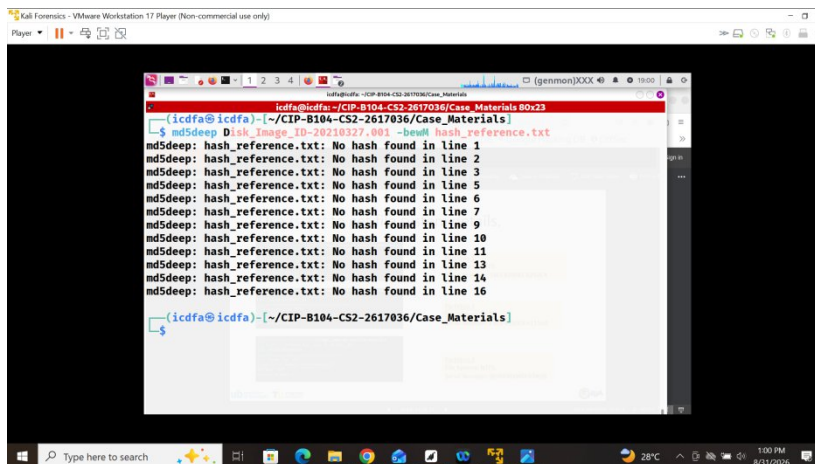
1. The fdisk -l result



```
icdffa@icdffa:~/CIP-B104-CS2-2617036/Case_Materials B0x23
icdffa@icdffa:~/CIP-B104-CS2-2617036/Case_Materials
$ fdisk -l
Disk Image_ID-20210327.001: 185 KiB, 189440 bytes, 370 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

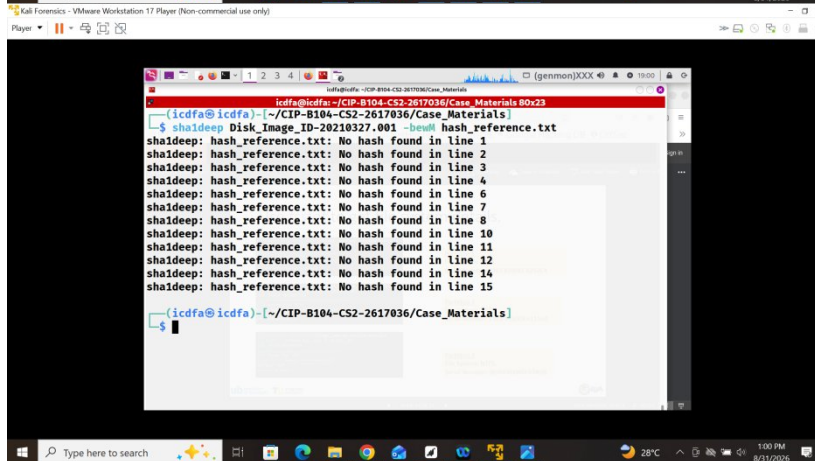
icdffa@icdffa:~/CIP-B104-CS2-2617036/Case_Materials$
```

2. Here I crossed reference the hash value with what I have in nano



```
icdffa@icdffa:~/CIP-B104-CS2-2617036/Case_Materials B0x23
icdffa@icdffa:~/CIP-B104-CS2-2617036/Case_Materials
$ md5deep Disk_Image_ID-20210327.001 -bom hash_reference.txt
md5deep: hash_reference.txt: No hash found in line 1
md5deep: hash_reference.txt: No hash found in line 2
md5deep: hash_reference.txt: No hash found in line 3
md5deep: hash_reference.txt: No hash found in line 5
md5deep: hash_reference.txt: No hash found in line 6
md5deep: hash_reference.txt: No hash found in line 7
md5deep: hash_reference.txt: No hash found in line 9
md5deep: hash_reference.txt: No hash found in line 10
md5deep: hash_reference.txt: No hash found in line 11
md5deep: hash_reference.txt: No hash found in line 13
md5deep: hash_reference.txt: No hash found in line 14
md5deep: hash_reference.txt: No hash found in line 16

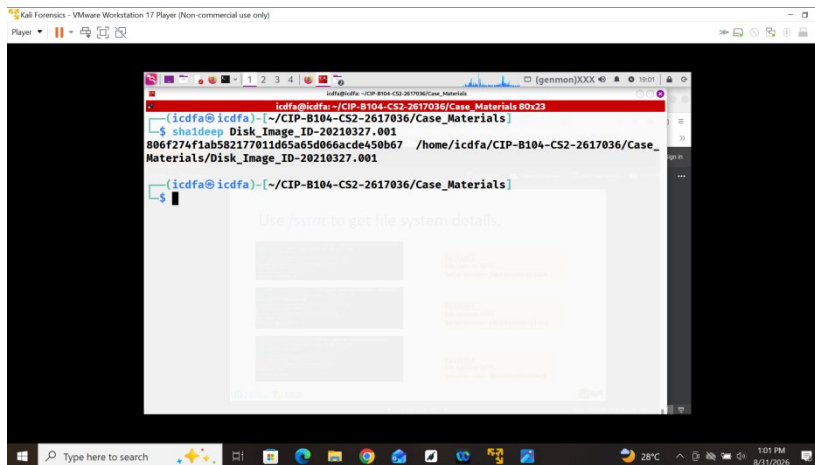
icdffa@icdffa:~/CIP-B104-CS2-2617036/Case_Materials$
```



```
icdffa@icdffa:~/CIP-B104-CS2-2617036/Case_Materials B0x23
icdffa@icdffa:~/CIP-B104-CS2-2617036/Case_Materials
$ sha1deep Disk_Image_ID-20210327.001 -bom hash_reference.txt
sha1deep: hash_reference.txt: No hash found in line 1
sha1deep: hash_reference.txt: No hash found in line 2
sha1deep: hash_reference.txt: No hash found in line 3
sha1deep: hash_reference.txt: No hash found in line 4
sha1deep: hash_reference.txt: No hash found in line 6
sha1deep: hash_reference.txt: No hash found in line 7
sha1deep: hash_reference.txt: No hash found in line 8
sha1deep: hash_reference.txt: No hash found in line 10
sha1deep: hash_reference.txt: No hash found in line 11
sha1deep: hash_reference.txt: No hash found in line 12
sha1deep: hash_reference.txt: No hash found in line 14
sha1deep: hash_reference.txt: No hash found in line 15

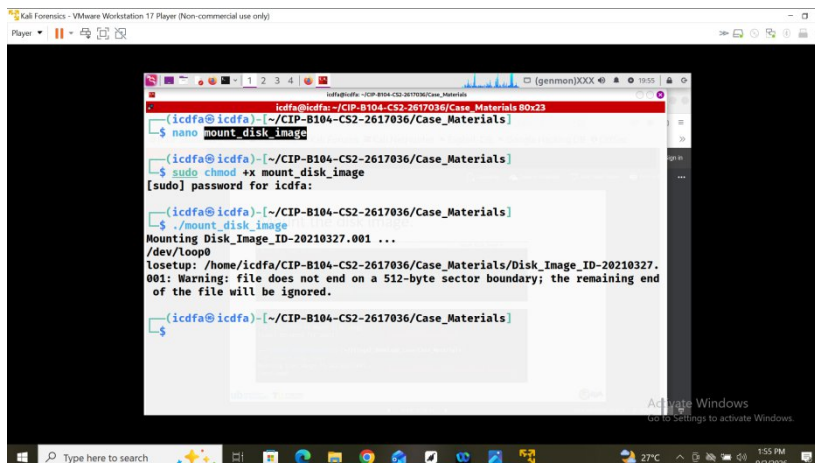
icdffa@icdffa:~/CIP-B104-CS2-2617036/Case_Materials$
```

3. The sha1deep value

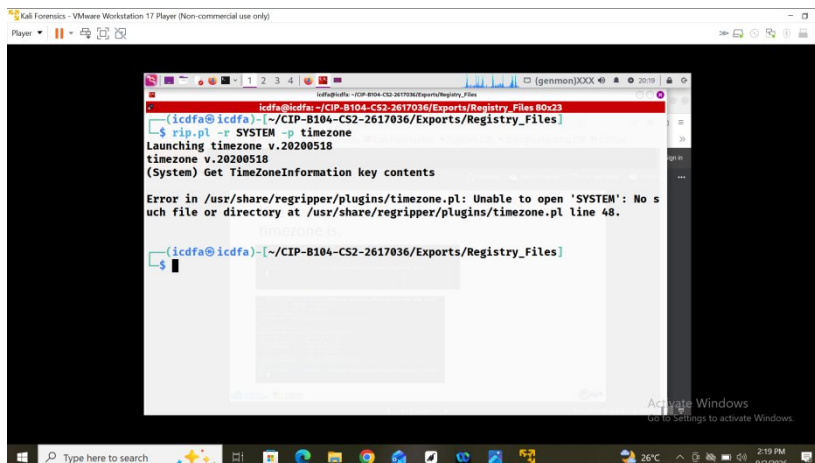


Data Leakage Registry and File

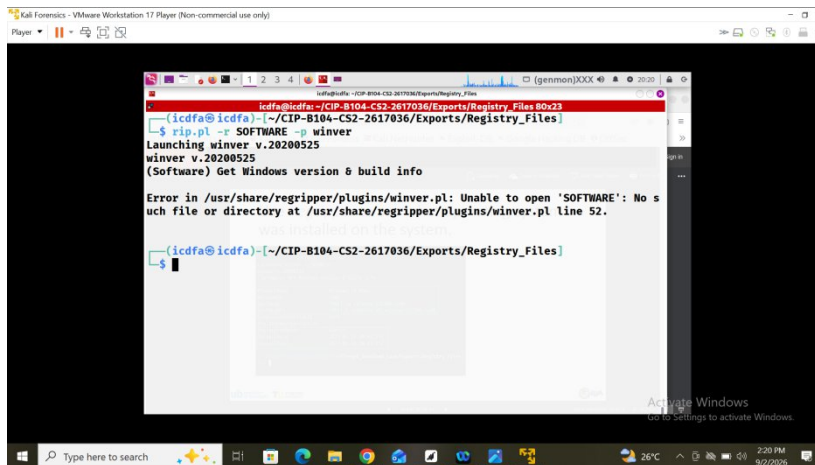
1. Mounting disk image with nano



2. Using regripper to find the operating system



3. Using regripper to find the system's nae



The screenshot shows a Kali Linux terminal window titled "Kali Forensics - VMware Workstation 17 Player (Non-commercial use only)". The terminal is running a command to use regripper on a Windows registry file. The output shows the tool launching winver, which reports the Windows version as 10.0.20200525. However, an error message is displayed: "Error in /usr/share/regripper/plugins/winver.pl: Unable to open 'SOFTWARE': No such file or directory at /usr/share/regripper/plugins/winver.pl line 52." The terminal prompt is now ready for the next command.

```
icdfe@icdfe:~/CIP-B104-CS2-2617036/Exports/Registry_Files$ rip.pl -r SOFTWARE -p winver
Launching winver v.20200525
winver v.20200525
(Software) Get Windows version & build info

Error in /usr/share/regripper/plugins/winver.pl: Unable to open 'SOFTWARE': No such file or directory at /usr/share/regripper/plugins/winver.pl line 52.

icdfe@icdfe:~/CIP-B104-CS2-2617036/Exports/Registry_Files$
```


Conclusion

The investigation was partially completed due to the forensic disk image being unavailable for download. Although this prevented full verification and attribution, relevant forensic analysis commands were successfully performed in the lab. Further conclusions regarding the presence, transfer, and distribution of the Beat Step files require access to the original disk image.